SOFTWARE REQUIREMENTS SPECIFICATION

I-Property: Rental Property Management System

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1. Related Document Register

This SRS is one artifact within a wider project document set. Each companion document is referenced in the relevant section of this SRS so a reader can locate the authoritative source for any topic.

Doc ID	Title	Ver	Status	Owner	SRS Sections
BRD-001	Business Requirements Document	1.3	Approved	Admin / IT-PMO	All sections, parent document
SAD-001	Software Architecture Document	TBD	Pending	IT Architecture	Section 5, Architecture Overview
SDD-001	Software Design Document	TBD	Pending	Development Lead	Section 6 (processing logic), Section 10 (field definitions)
UI-001	UI/UX Design Specification and Wireframes	TBD	Pending	UX Designer	Section 9, Screen Inventory and UI Requirements
IDD-001	Integration Design Document (Oracle Financial)	TBD	Pending	Integration Architect	Section 8, API and Integration Specifications
TP-001	Test Plan and Test Cases	TBD	Pending	QA Lead	Section 13 RTM; verification steps in Sections 6 and 7
DP-001	Deployment and Infrastructure Plan	TBD	Pending	IT Infrastructure	Section 11, NFR verification methods
TRM-001	Full Requirements Traceability Matrix	TBD	Pending	BA / PMO	Section 13, RTM preview

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3. Introduction

3.1 Purpose

This document specifies the software requirements for the I-Property system at a level of detail sufficient for a development team to implement, a QA team to write test cases, and stakeholders to verify that the delivered system meets the stated business needs. Every functional requirement includes exact inputs, processing logic, outputs, error conditions, and step-by-step verification procedures.

3.2 Intended Audience

Audience	How They Use This Document
Solution Architects	Sections 4, 7, 10: architecture decisions, API contracts, and non-functional requirements
Backend Developers	Section 5: functional requirement processing logic and error codes; Section 7: API specifications; Section 9: field definitions
Frontend Developers	Section 8: screen inventory and UI requirements; Section 9: field validation rules
QA and Test Engineers	Verification steps within each FR block in Sections 5 and 6; Section 12: RTM
Business Stakeholders	Sections 5 and 6: requirement descriptions and expected outputs for sign-off validation
Integration Team	Section 7: Oracle Financial API contracts, payload schemas, and error handling
Database Architects	Section 9: field definitions and data types, elaborated in SDD-001 Section 4

3.3 Scope

- Phase 1 includes new request, renewal, and vacated or termination workflows with full multi-level approval routing.
- Phase 1 includes utility bills payment tracking and ABH approval.
- Phase 1 includes automated monthly payment summaries, Oracle Financial integration, and the notification engine.
- Phase 1 includes role-based access control across 7 roles, audit trail, and document management.
- Phase 2 features are out of scope for this SRS and include the signage module, Capex integration, and cheque cancellation.
- Payment disbursement execution is always out of scope as it is handled entirely within Oracle Financial.

3.4 Conventions

- All timestamps are stored in UTC and displayed in Pacific Time (UTC-8) in the user interface.
- All monetary amounts are in USD unless stated otherwise.
- API base URL: https://iproperty.internal/api/v1 (to be confirmed in IDD-001).
- HTTP status codes follow RFC 7231.
- All API requests require an Authorization Bearer JWT header.
- Field lengths refer to string character counts or numeric precision expressed as scale and precision.

4. System Overview

4.1 System Context

I-Property is a multi-role web application deployed on the organization's internal infrastructure. It does not execute payments. It generates payment requests and transmits them to Oracle Financial. The full system context diagram is maintained in SAD-001 Section 3.

4.2 User Roles and Permissions Summary

Role	System Access	Cannot Do
Area Officer	Create new, renewal, and vacated requests; enter utility bills; view own requests only.	Approve any request; view other areas; access Finance screens.
Area Admin	All Area Officer permissions; access Print Queue; initiate vacated requests.	Approve requests; view payment amounts.
ABH / MR / ARH	Approve or reject requests; approve monthly payment summaries; stop payments; view regional reports.	Create requests; access User Management.
Director Office	Approve or reject requests escalated from ABH; view request history.	Create requests; view payment data.
Admin Head Office	Review and complete requests; edit agreement details; upload vacated notices.	Approve as ABH or Director; manage users.
Finance Department	View and export payment reports; mark stop-payment notes; read-only access to agreements.	Create or approve requests; edit agreement data.
System Administrator	Full access including user management, role assignment, notification configuration, and audit log viewer.	Approve business requests as no business role is assigned.

5. System Architecture Overview

The full architectural specification is maintained in SAD-001. This section provides the context required to understand the system behavior described in later sections.

5.1 Tier Structure

Tier	Technology (confirmed in SAD-001)	Key Responsibility
Presentation	React SPA served over HTTPS; responsive desktop-first layout defined in UI-001	All screens listed in Section 9; form validation on the client side as a duplicate of server-side validation

Tier	Technology (confirmed in SAD-001)	Key Responsibility
Application	RESTful API layer (Node.js or Java Spring, to be confirmed in SAD-001)	Business logic, workflow engine, approval routing, notification queue, and Oracle Financial integration adapters
Data	Relational RDBMS (PostgreSQL or Oracle, to be confirmed in SAD-001)	All entities in Section 9; audit log table (append-only); document path references
Document Store	Network file share or S3-compatible object store (to be confirmed in DP-001)	Binary storage for uploaded files; the application stores the path reference in the database only
Notification	Async job queue feeding SMTP relay (Redis-backed, to be confirmed in SAD-001)	Decoupled notification delivery with retry on failure; prevents blocking of approval transactions

5.2 Authentication and Session

- Authentication is handled via organization SSO or LDAP using OAuth 2.0 or SAML 2.0 (to be confirmed in SAD-001 Section 5).
- A JWT is issued on successful login with an 8-hour expiry. The refresh token expires after 24 hours.
- Session inactivity timeout is set to 30 minutes. The UI displays a 2-minute warning modal before auto-logout.
- Accounts are locked after 5 consecutive failed login attempts. Unlock is performed by the System Administrator or via a 30-minute auto-reset.
- All API endpoints validate the JWT on every request. A 401 is returned if the token is absent, expired, or invalid.

5.3 Audit Architecture

- A separate audit_log table contains the following columns: log_id, table_name, record_id, action (INSERT, UPDATE, or DELETE), old_value as JSON, new_value as JSON, user_id, ip_address, and timestamp_utc.
- Database triggers or application-layer interceptors write to audit_log on every create, update, or delete operation.
- No UPDATE or DELETE is permitted on the audit_log table. This is enforced at the database role level.
- Audit logs are retained for 3 years and then archived to cold storage. They remain readable via the Audit Log Viewer, which is accessible to the System Administrator only.

6. Functional Requirements

Each requirement is specified with a description, pre-conditions, trigger, inputs, processing logic including business rules and calculations, expected outputs, error handling with specific error codes and messages, and step-by-step verification procedures for QA.

6.1 Request Initiation and Reference ID Generation

FR-NR-01 | Request Submission and Reference ID Generation | Must Have | BRD: FR-NR-01, FR-NR-02	
Description	The Area Officer completes and submits a new rental request form. The system validates all inputs, persists the record, generates a globally unique reference ID, and routes the request to the appropriate ABH based on the area selected.
Pre-conditions	User is authenticated with role set to Area Officer. The user's assigned area is configured in the system. At least one active Landlord Profile and one active Property Profile exist.
Trigger	The user clicks the Submit button on the New Rental Request form (SCR-03).
Inputs	landlord_id: string, required. Must match an active record in the landlord_profile table. property_id: string, required. Must match an active record in the property_profile table linked to the landlord_id. area_code: string, required. Must match a configured area; must match the user's assigned area. agreement_start_date: date in DD/MM/YYYY format, required. Must not be in the past. agreement_end_date: date in DD/MM/YYYY format, required. Must be after the agreement_start_date. rental_amount: decimal(15,2), required. Range: 1.00 to 99,999,999.99. tax_rate: decimal(5,2), required. Range: 0.00 to 100.00. agreement_type: enum of New, Renewal, or Transfer, required. document_attachments: array of files, optional. PDF, JPG, or PNG; maximum 10 MB per file; maximum 5 files.
Processing Logic	1. Server-side validation executes for all fields. Client-side validation is a UX duplicate only. 2. If any validation fails: return HTTP 422 with an error array. Do not persist any data. 3. If all validations pass: open a database transaction. 4. Insert a record into the requests table with status set to PENDING_ABH. 5. Generate reference_id in the format REQ-{YYYY}-{6-digit zero-padded sequence}, for example REQ-2025-000142. The sequence resets each calendar year. A database sequence object is used to guarantee uniqueness under concurrent submissions. 6. Look up the ABH user IDs for the submitted area_code from the area_abh_mapping table. 7. Commit the transaction. 8. Enqueue notification job NE-01 asynchronously. This must not block the HTTP response. 9. Return HTTP 201.
Outputs	HTTP 201 Created with request_id, status, created_at timestamp, and assigned_abh_ids in the response body. The UI displays a success banner with the reference ID. The form is cleared and locked. A new row is created in the requests table and the request_audit_trail table. NE-01 is dispatched to ABH within 2 minutes.
Error Handling	ERR-NR-01-001: landlord_id not found. HTTP 422. Message: Landlord ID does not exist or is inactive. ERR-NR-01-002: property_id not linked to landlord_id. HTTP 422. Message: Property is not associated with the selected Landlord.

	ERR-NR-01-003: area_code not in user's allowed areas. HTTP 403. Message: You do not have permission to submit requests for this area. ERR-NR-01-004: agreement_start_date in the past. HTTP 422. Message: Start date cannot be in the past. ERR-NR-01-005: agreement_end_date on or before start_date. HTTP 422. Message: End date must be after Start date. ERR-NR-01-006: rental_amount out of range. HTTP 422. Message: Amount must be between 1.00 and 99,999,999.99. ERR-NR-01-007: file type not allowed. HTTP 422. Message: Only PDF, JPG, and PNG files are permitted. ERR-NR-01-008: file size exceeds 10 MB. HTTP 422. Message: File {filename} exceeds the 10 MB limit. ERR-NR-01-009: database transaction failure. HTTP 500. Message: Request could not be saved. Please try again. If the issue persists, contact IT Support. The full error is logged server-side. No stack trace is exposed to the client.
Verification Steps	TC-001-A: Submit a fully valid request. Verify HTTP 201, reference ID format REQ-YYYY-NNNNNN, a DB record with status PENDING_ABH, and NE-01 email received by ABH within 2 minutes. TC-001-B: Submit with an invalid landlord_id. Verify HTTP 422, error code ERR-NR-01-001, and no DB record created. TC-001-C: Submit with start_date set to yesterday. Verify HTTP 422 and ERR-NR-01-004. TC-001-D: Simulate 50 concurrent submissions. Verify all receive unique reference IDs with no duplicates. TC-001-E: Submit with an 11 MB file. Verify HTTP 422, ERR-NR-01-008, and that the file is not stored on disk.

6.2 Approval Routing and Status Transitions

FR-NR-02 | ABH Approve or Reject | Must Have | BRD: FR-NR-03, FR-NR-04	
Description	ABH reviews a pending request and takes an Approve or Reject action. The system enforces mandatory remarks on rejection, transitions the request status, and routes or closes the request accordingly.
Pre-conditions	The request exists with status set to PENDING_ABH. The authenticated user has role ABH, MR, or ARH and is mapped to the request's area_code.
Trigger	ABH clicks Approve or Reject on the Approval Queue screen (SCR-04) or the Request Detail screen (SCR-05).
Inputs	request_id: string, required. action: enum of APPROVE or REJECT, required. remarks: string, required when action is REJECT. Minimum 1 character, maximum 500 characters.
Processing Logic	1. Validate that the request status is PENDING_ABH. If not, return ERR-AP-02-001. 2. Validate that the acting user is the mapped ABH for the request's area. If not, return HTTP 403.

	3. If action is APPROVE: update requests.status to PENDING_DIRECTOR. Insert a row in approval_history with approver_id, role, action set to APPROVED, and remarks set to null. 4. If action is REJECT: validate that remarks are not empty. If empty, return ERR-AP-02-002. Update requests.status to REJECTED. Insert a row in approval_history with action set to REJECTED and the provided remarks. No further routing occurs. 5. Commit the transaction and enqueue the appropriate notification. NE-02 for approve, NE-03 for reject.
Outputs	HTTP 200 OK. For approve: response body includes request_id and new_status as PENDING_DIRECTOR. For reject: response body includes request_id, new_status as REJECTED, and remarks. The request is removed from ABH's Approval Queue. The status badge is updated on the Request Detail screen.
Error Handling	ERR-AP-02-001: Request not in PENDING_ABH status. HTTP 409. Message: This request has already been actioned. ERR-AP-02-002: Reject attempted without remarks. HTTP 422. Message: Rejection remarks are mandatory. ERR-AP-02-003: Acting user not mapped to the request area. HTTP 403. Message: You are not authorised to action requests for this area.
Verification Steps	TC-002-A: Approve a valid PENDING_ABH request. Verify status is PENDING_DIRECTOR, an approval_history row exists, and NE-02 email is sent to Director within 2 minutes. TC-002-B: Reject without remarks. Verify HTTP 422, ERR-AP-02-002, and status unchanged in the DB. TC-002-C: Reject with valid remarks. Verify status is REJECTED, an approval_history row exists with remarks, and NE-03 email is sent to Area Officer. TC-002-D: ABH from Area B attempts to action an Area A request. Verify HTTP 403. TC-002-E: Attempt to approve an already-approved request with status PENDING_DIRECTOR. Verify HTTP 409 and ERR-AP-02-001.

FR-NR-03 | Director and Admin HO Approval, Full Chain Completion | Must Have | BRD: FR-NR-04, FR-NR-05, FR-NR-06	
Description	The Director Office and Admin Head Office complete the remaining approval chain. The Admin Head Office marks the request as Completed, which triggers simultaneous notifications to Area Admin and Finance.
Pre-conditions	For Director action: request status is PENDING_DIRECTOR and the user role is Director Office. For Admin HO action: request status is PENDING_ADMIN_HO and the user role is Admin Head Office.
Trigger	The approver clicks Approve or Reject on SCR-04 or SCR-05. The Admin HO clicks Mark as Completed.
Inputs	request_id, action (APPROVE, REJECT, or COMPLETE), and remarks which are mandatory on REJECT.
Processing Logic	Director APPROVE: status transitions to PENDING_ADMIN_HO. NE-04 is sent to Admin Head Office.

	Director REJECT: status transitions to REJECTED. NE-05 is sent to ABH and Area Officer. Admin HO COMPLETE: status transitions to COMPLETED. The Approval Summary PDF is generated (see FR-NR-04). NE-06 is enqueued to both Area Admin and Finance simultaneously. All transitions insert a row in approval_history and commit before enqueuing notifications.
Outputs	HTTP 200 with new_status in the response body for each action. On COMPLETE, the approval_summary_url is returned in the response body pointing to the generated PDF.
Error Handling	Same error pattern as FR-NR-02 with status-appropriate codes. ERR-AP-03-001: Admin HO attempts Complete when status is not PENDING_ADMIN_HO. HTTP 409.
Verification Steps	TC-003-A: Full happy path. Submit, ABH approves, Director approves, Admin HO completes. Verify all status transitions in the DB, all 4 notifications received, and approval_summary_url is accessible. TC-003-B: Director rejects. Verify NE-05 is sent to both ABH and Area Officer. TC-003-C: Finance notification body must contain agreement_id, landlord_name, rental_amount, tax_rate, and effective_date. Verify email content against the NE-06 template.

6.3 Approval Summary PDF Generation

FR-NR-04 | Auto-Generate Approval Summary PDF | Must Have | BRD: FR-NR-08	
Description	When the Admin HO marks a request as Completed, the system auto-generates an Approval Summary PDF listing every approver in chain order with name, role, action taken, and timestamp. This PDF is appended to the print package.
Pre-conditions	The request status has just transitioned to COMPLETED. All approval_history rows for the request exist in the database.
Trigger	Triggered automatically by the COMPLETE action in FR-NR-03. This is not a user-initiated action.
Inputs	Internally derived from the approval_history table: approver_name, role, action, remarks if any, and timestamp_utc converted to PST.
Processing Logic	1. Query approval_history for the request_id ordered by timestamp ascending. 2. Render a PDF template containing: request reference_id, property address, landlord name, and a table of approvers with sequence, name, role, action, date and time in PST, and remarks. 3. Store the PDF in the document store and insert a reference in request_documents with doc_type set to APPROVAL_SUMMARY. 4. Return the document URL in the COMPLETE response body. 5. The PDF must be generated within 10 seconds of the COMPLETE action.
Outputs	PDF file accessible at the returned URL. Filename format: APPROVAL_SUMMARY_{request_id}.pdf. PDF contains a header with document title and reference ID, the approver table with all chain members, and a footer with the generation timestamp. The document is listed in the Print Queue (SCR-11) alongside the agreement documents.

Error Handling	ERR-PDF-04-001: PDF generation exceeds 10 seconds. HTTP 202 Accepted. Response includes pdf_status set to GENERATING and a poll_url. The UI shows a spinner and polls every 2 seconds. ERR-PDF-04-002: PDF generation failure. The error is logged. pdf_status is set to FAILED. The System Administrator is alerted. Area Admin is shown a message directing them to contact IT Support with the log reference ID.
Verification Steps	TC-004-A: Complete a request with a 3-level approval chain. Verify the PDF is generated within 10 seconds, contains all 3 approvers with correct names and timestamps, and is accessible via the returned URL. TC-004-B: Verify the PDF timestamp is displayed in PST (UTC-8). TC-004-C: Simulate a PDF engine timeout. Verify HTTP 202, poll_url in the response, and that the UI shows the generating state before the PDF becomes available.

6.4 Form Abandonment and Unsaved Changes

FR-NR-05 | Discard Unsaved Changes on Navigation Away | Must Have | BRD: FR-NR-12	
Description	If a user navigates away from a partially completed form without saving, the system prompts for confirmation. If confirmed, all entered data is discarded. No partial record is persisted to the database.
Pre-conditions	The user has opened any create or edit form on SCR-03, SCR-07, or SCR-08. The user has modified at least one field without submitting.
Trigger	The user triggers browser Back, closes the tab, navigates to another screen via the menu, or clicks Cancel.
Inputs	Navigation intent event from browser beforeunload or the in-app router guard.
Processing Logic	1. Intercept navigation using the browser beforeunload event and the in-app router guard. 2. If the form is dirty (any field changed from its initial state): display a modal with title Unsaved Changes and buttons Stay on Page and Discard Changes. 3. If the user clicks Stay on Page: dismiss the modal. The user remains on the current form with no data loss. 4. If the user clicks Discard Changes: clear the form state, reset to initial values, and proceed with navigation. 5. If the form has not been modified: allow navigation without a prompt. 6. On browser tab close: the browser's native beforeunload fires. Custom message display varies by browser and is an accepted limitation.
Outputs	Modal is displayed within 100ms of the navigation event. No data is written to the database until Submit is clicked. On discard: the form resets to empty for a new form or to original values for an edit form.
Error Handling	No specific error codes apply. The failure mode is a modal that does not appear, which risks accidental data loss. Mitigation: automated regression tests in TP-001 must cover this scenario across Chrome, Firefox, and Edge.
Verification Steps	TC-005-A: Fill 3 fields on the New Request form and click browser Back. Verify the modal appears. TC-005-B: Fill fields and click Discard. Verify the form clears and no DB record exists for that session.

	TC-005-C: Fill fields and click Stay on Page. Verify all entered data is preserved. TC-005-D: Open the form without modifying any field and click Back. Verify no modal appears (form not dirty). TC-005-E: Repeat TC-005-A in Firefox and Edge.

6.5 Monthly Payment Summary

FR-NR-06 | Auto-Generate and Approve Monthly Payment Summary | Must Have | BRD: FR-NR-09, FR-NR-10	
Description	On the last working day of each month, the system auto-generates a payment summary covering all active agreements. ABH or GM reviews and approves it with the option to stop payment for specific landlords. Finance receives the approved report.
Pre-conditions	Active agreements exist with status COMPLETED and payment not stopped. The system scheduler is operational and a working day calendar is configured.
Trigger	Automated scheduler trigger at 08:00 PST on the last working day of the month.
Inputs	Internally derived from all active agreements including rental amounts, tax rates, and bank details. ABH input: stop_payment flag (boolean) and stop_reason (string, minimum 10 characters when flag is true).
Processing Logic	Scheduler step 1: Query all agreements with status ACTIVE and payment_stopped set to false. Scheduler step 2: For each agreement, calculate gross_amount, tax_deduction as rental_amount multiplied by tax_rate divided by 100, and net_payable as gross_amount minus tax_deduction. Scheduler step 3: Insert the batch into the monthly_payment_summary table with period set to the current month and summary_status set to PENDING_ABH. Scheduler step 4: Enqueue NE-07 to all ABH and GM users. ABH review step 5: ABH may set payment_stopped to true on any row with a mandatory stop_reason. ABH review step 6: ABH clicks Approve Summary. summary_status transitions to APPROVED. NE-08 is enqueued to Finance. Step 7: The report is locked. No further edits are permitted once status is APPROVED.
Outputs	Scheduler sends NE-07 to ABH and GM by 08:05 PST on the generation date. Finance receives NE-08 with a download link to the approved report. The report is exportable as Excel and PDF. Stopped payments appear in the report with status STOPPED and the stop_reason column populated.
Error Handling	ERR-PS-06-001: Scheduler fails to run. System Admin receives an automated alert. A manual trigger endpoint is available at POST /api/v1/payment-summary/generate with admin authentication. ERR-PS-06-002: Stop payment attempted without stop_reason. HTTP 422. Message: A reason is required to stop a payment (minimum 10 characters). ERR-PS-06-003: Attempt to edit an approved summary. HTTP 409. Message: This summary has been approved and is locked.

Verification Steps	TC-006-A: Trigger the scheduler via the admin endpoint in the test environment. Verify the summary is generated for all active agreements and NE-07 is received. TC-006-B: Set stop_payment without a reason. Verify HTTP 422 and ERR-PS-06-002. TC-006-C: Set stop_payment with a valid reason. Verify payment_stopped is true in the DB and the stopped entry appears in the Excel export with the reason. TC-006-D: Approve the summary. Verify summary_status is APPROVED, NE-08 is sent to Finance, and a subsequent edit attempt returns HTTP 409. TC-006-E: Verify the Excel export. Each row must contain landlord_id, landlord_name, agreement_id, period, gross_amount, tax_deduction, net_payable, masked bank account (last 4 digits visible), and status.

6.6 Vacated and Termination, Stop Payment

FR-VT-01 | Automated Stop-Payment on Vacated Approval | Must Have | BRD: FR-VT-01, FR-VT-02	
Description	When the Director approves a vacated or termination request, the system immediately halts payment generation for that agreement and notifies Finance with precise stop-payment details. No manual intervention is required.
Pre-conditions	A vacated request exists with status PENDING_DIRECTOR. The agreement linked to the request has status ACTIVE.
Trigger	The Director clicks Approve on a vacated request.
Inputs	request_id, action set to APPROVE, and termination_date as captured at request creation.
Processing Logic	1. Update requests.status to PENDING_ADMIN_HO as part of the standard chain. Simultaneously set agreements.payment_stopped to true, agreements.payment_stop_date to termination_date, and agreements.stop_reason to VACATED. 2. Remove the agreement from future payment scheduler runs immediately. The scheduler checks the payment_stopped flag. 3. Enqueue NE-10 as urgent to Finance, including agreement_id, landlord_id, landlord_name, last_payment_date, and termination_date. 4. Insert an audit_log entry with action set to PAYMENT_STOPPED, triggered by vacated approval. 5. All steps above are executed within a single database transaction. If any step fails, all changes are rolled back.
Outputs	HTTP 200. Finance NE-10 email is dispatched within 2 minutes. The agreement is excluded from the next monthly_payment_summary generation run. An audit log entry is verifiable by the System Administrator.
Error Handling	ERR-VT-01-001: Database transaction rollback. HTTP 500. Finance is not notified as the notification is only sent on a successful commit. System Admin is alerted and manual intervention is required as documented in DP-001 Section 6. If Finance notification delivery fails due to an SMTP error: the notification_log records status as FAILED. Retry occurs 3 times at 5-minute intervals. If all retries fail, the System Administrator is alerted with the log_id.

Verification Steps	TC-007-A: Approve a vacated request. Verify agreement.payment_stopped is true in the DB, payment_stop_date equals termination_date, and NE-10 is received by Finance within 2 minutes. TC-007-B: Run the payment scheduler after TC-007-A. Verify the vacated agreement is not included in the generated summary. TC-007-C: Verify the NE-10 email body contains all required fields: agreement_id, landlord_id, landlord_name, last_payment_date, and termination_date. TC-007-D: Simulate a database failure during the transaction. Verify rollback, agreement.payment_stopped remains false, Finance is not notified, and the System Admin alert is triggered.

6.7 Utility Bill Submission and Duplicate Prevention

FR-UT-01 | Utility Bill Entry with Duplicate Check | Must Have | BRD: FR-UT-01 to FR-UT-04	
Description	The Area Officer submits a utility bill against an active agreement. The system validates all inputs, checks for duplicate entries based on agreement, billing period, and bill type, routes the bill to ABH for approval, and notifies Finance upon ABH approval.
Pre-conditions	An agreement with utility connections exists with status ACTIVE. The user has role Area Officer with access to the agreement's area.
Trigger	The user completes the Utility Bill form (SCR-08) and clicks Submit.
Inputs	agreement_id: string, required. Must reference an ACTIVE agreement. bill_type: enum of ELECTRICITY, GAS, WATER, or OTHER, required. billing_period: string in MM/YYYY format, required. Must not be in the future. units_consumed: decimal(10,2), required. Minimum 0. bill_amount: decimal(15,2), required. Minimum 0.01. advance_tax: decimal(15,2), optional. Default is 0, minimum 0. bill_document: file, required. PDF, JPG, or PNG; maximum 10 MB.
Processing Logic	1. Validate all fields per the rules above. 2. Duplicate check: query utility_bills for any record matching agreement_id, bill_type, and billing_period where status is not REJECTED. If a match exists, return ERR-UT-01-003. 3. Calculate net_payable as bill_amount plus advance_tax. This is stored as a derived field. 4. Store the uploaded file in the document store and record the path in utility_bills.document_path. 5. Insert a utility_bills row with status set to PENDING_ABH. 6. Enqueue NE-11 to the ABH for the agreement's area.
Outputs	HTTP 201 Created with utility_bill_id in format UTIL-YYYY-NNNNNN, status as PENDING_ABH, and net_payable in the response body. NE-11 is sent to ABH within 2 minutes.
Error Handling	ERR-UT-01-001: agreement_id is not ACTIVE. HTTP 422. Message: Agreement is not active or does not exist. ERR-UT-01-002: billing_period is in the future. HTTP 422. Message: Billing period cannot be in the future.

	ERR-UT-01-003: Duplicate bill detected. HTTP 409. Message: A bill for this type and period already exists for this agreement. To correct it, contact your Admin. ERR-UT-01-004: bill_amount is 0. HTTP 422. Message: Bill amount must be greater than zero.
Verification Steps	TC-008-A: Submit a valid utility bill. Verify HTTP 201, UTIL-YYYY-NNNNNN format, a DB row, and NE-11 received within 2 minutes. TC-008-B: Submit the same agreement_id, bill_type, and billing_period twice. The second attempt must return HTTP 409 and ERR-UT-01-003. TC-008-C: ABH approves the utility bill. Verify NE-12 is sent to Finance with bill_id, net_payable, and payment due date. TC-008-D: Submit with a 10.1 MB file. Verify HTTP 422 and that the file is not stored on disk. TC-008-E: Verify net_payable calculation: bill_amount of 10000 plus advance_tax of 2500 must equal net_payable of 12500.00 in the DB.

7. System Workflows

The tables below define step-by-step workflows with actor, action, system response, status transition, and notification triggered at each step. These are the direct input for sequence diagrams in SDD-001.

7.1 New Rental Request, End-to-End

Step	Actor	Action	System Response	Status After	Notification
1	Area Officer	Fills and submits New Request form	Validates inputs, generates REQ-YYYY-NNNNNN, persists record	PENDING_ABH	NE-01 to ABH within 2 min
2a	ABH	Approves request	Updates status, logs approval_history row	PENDING_DIRECTOR	NE-02 to Director within 2 min
2b	ABH	Rejects with mandatory remarks	Updates status, logs rejection with remarks, request closed	REJECTED	NE-03 to Area Officer within 2 min
3a	Director	Approves request	Updates status, logs approval_history row	PENDING_ADMIN_HO	NE-04 to Admin HO within 2 min
3b	Director	Rejects with mandatory remarks	Updates status, logs rejection, request closed	REJECTED	NE-05 to ABH and Area Officer within 2 min
4	Admin HO	Reviews documents, inputs details, clicks Complete	Updates status, generates Approval Summary PDF, notifies 2 parties	COMPLETED	NE-06 to Area Admin and Finance simultaneously within 2 min
5	System	Auto-generates Approval Summary PDF	PDF stored, URL available in Print Queue (SCR-11)	COMPLETED	None

Step	Actor	Action	System Response	Status After	Notification
6	Area Admin	Downloads and prints agreement package from SCR-11	Serves agreement and approval summary as a combined PDF package	COMPLETED	None
7	Finance	Processes payment in Oracle Financial using NE-06	Out of scope in i-Property, handled in Oracle Financial	COMPLETED	None

7.2 Renewal Request Workflow

Step	Actor	Action	System Response	Status After	Notification
1	Admin	Views expiry report (SCR-06), identifies expiring agreements	Report shows agreements expiring within the look-ahead window (default 90 days)	N/A	None
2	Area Office	Selects agreement, clicks Initiate Renewal	Pre-populates form with existing data, reserves new reference ID	DRAFT	None
3	ABH / MR / ARH	Edits details, adds remarks, submits renewal	Generates new REQ-YYYY-NNNNNN, cross-references original_request_id	PENDING_ABH	NE-01 to ABH within 2 min
4-6	All Approvers	Same chain as New Request Steps 2a through 4	Identical to Section 6.1	COMPLETED	As per Section 6.1
7a	System	If rejected at any stage	Status set to CLOSED, no further actions permitted	CLOSED	NE-03 or NE-05 as applicable

7.3 Vacated and Termination Workflow

Step	Actor	Action	System Response	Status After	Notification
1	Area Admin	Opens Vacated Request form, enters termination date and reason,	Validates termination date is not in the past, stores form data as DRAFT	DRAFT	None

Step	Actor	Action	System Response	Status After	Notification
		uploads documents			
2	System	On submission	Generates new reference ID, routes to ABH	PENDING_ABH	NE-01 to ABH within 2 min
3-4	ABH / Director	Standard approval chain	Same as Steps 2a through 3a of Section 6.1	PENDING_ADMIN_HO	NE-02, NE-04
5	System	On Director APPROVE	Sets agreement.payment_stopped to true, enqueues urgent NE-10 to Finance	PENDING_ADMIN_HO	NE-10 to Finance, urgent, within 2 min
6	System	Payment scheduler next run	Agreement excluded from monthly_payment_summary generation	None	None
7	Admin HO	Issues vacated notice, uploads scanned copy	Document stored and linked to request, status transitions to COMPLETED	COMPLETED	None

7.4 Utility Bill Payment Workflow

Step	Actor	Action	System Response	Status After	Notification
1	Area Officer	Searches by Agreement ID or Landlord ID on SCR-08	Returns matching records within 2 seconds and shows the utility connection list	None	None
2	Area Officer	Selects connection, chooses bill type, fills bill details, uploads document	Duplicate check runs. If passed: stores bill. If failed: ERR-UT-01-003 returned.	PENDING_ABH	NE-11 to ABH within 2 min
3	ABH	Reviews and approves utility bill	Updates status, enqueues NE-12 to Finance	APPROVED	NE-12 to Finance within 2 min
4	Finance	Processes utility payment in Oracle Financial	Out of scope in i-Property	None	None

8. API Specifications

All endpoints are prefixed with: https://iproperty.internal/api/v1

All requests require an Authorization Bearer JWT header. Content-Type is application/json except for file upload endpoints which use multipart/form-data.

Full API contracts including all field-level schemas, pagination, and filtering parameters are maintained in IDD-001. This section covers the primary endpoints referenced by the functional requirements.

8.1 Request Management Endpoints

POST /requests | Submit new rental request	
Request Headers	Authorization: Bearer {JWT} Content-Type: multipart/form-data
Request Body	{ "landlord_id": "LL-000142", "property_id": "PROP-000089", "area_code": "AREA-EAST", "agreement_start_date": "01/05/2025", "agreement_end_date": "30/04/2026", "rental_amount": 75000.00, "tax_rate": 15.00, "agreement_type": "New", "remarks": "Ground floor office unit" } Plus files[]: document_attachments (multipart)
201 Response	HTTP 201 Created { "request_id": "REQ-2025-000142", "status": "PENDING_ABH", "created_at": "2025-05-01T03:32:11Z", "assigned_abh_ids": ["USR-0045"] }
Error Responses	400 Bad Request: malformed JSON. 401 Unauthorized: missing or expired JWT. 403 Forbidden: user area mismatch (ERR-NR-01-003). 422 Unprocessable Entity: field validation failures with error array including field, code, and message. 500 Internal Server Error: database failure (ERR-NR-01-009). Stack trace logged server-side only.
Notes	Idempotency: duplicate submission of an identical payload within 60 seconds returns the existing request_id with HTTP 200. File uploads are stored before the DB insert. If the DB insert fails, uploaded files are cleaned up.

PATCH /requests/{request_id}/action | Approve or reject a request at any chain stage	
Request Headers	Authorization: Bearer {JWT} Content-Type: application/json
Request Body	{ "action": "REJECT", "remarks": "Rental amount exceeds approved budget ceiling for this area" } Valid action values: APPROVE, REJECT, or COMPLETE (Admin HO only).
200 Response	HTTP 200 OK { "request_id": "REQ-2025-000142", "new_status": "REJECTED", "actioned_by": "USR-0045", "actioned_at": "2025-05-02T09:14:33Z" }
Error Responses	401 Unauthorized.

	403 Forbidden: user not permitted to action this request. 409 Conflict: request not in expected status (ERR-AP-02-001), includes current_status in response. 422 Unprocessable: REJECT submitted without remarks (ERR-AP-02-002).
Notes	Optimistic locking: the request row includes a version integer. The PATCH must include the current version to prevent concurrent approval conflicts. On version mismatch: HTTP 409 with message directing the user to refresh and retry.

GET /requests/{request_id} | Retrieve full request details including approval history	
Request Headers	Authorization: Bearer {JWT}
Request Body	No body. The request_id is passed in the path.
200 Response	HTTP 200 OK { "request_id": "REQ-2025-000142", "status": "PENDING_DIRECTOR", "landlord": { "id": "LL-000142", "name": "Ahmad Raza" }, "property": { "id": "PROP-000089", "address": "Shop 5, Block B" }, "rental_amount": 75000.00, "tax_rate": 15.00, "approval_history": [{ "stage": 1, "role": "ABH", "actor": "Bilal Khan", "action": "APPROVED", "timestamp": "2025-05-02T07:11:00Z", "remarks": null }], "documents": [{ "doc_type": "AGREEMENT", "url": "https://..." }] }
Error Responses	401 Unauthorized. 403 Forbidden: user not permitted to view this request. 404 Not Found.
Notes	Response is filtered by role. Finance can view agreement financials but not landlord bank details. Area Officers cannot view requests from other areas.

8.2 Utility Bill Endpoints

POST /utility-bills | Submit a utility bill for approval	
Request Headers	Authorization: Bearer {JWT} Content-Type: multipart/form-data
Request Body	{ "agreement_id": "AGR-2025-000031", "bill_type": "ELECTRICITY", "billing_period": "04/2025", "units_consumed": 340.50, "bill_amount": 10200.00, "advance_tax": 1530.00 } Plus file: bill_document (multipart, required).
201 Response	HTTP 201 Created { "utility_bill_id": "UTIL-2025-000089", "status": "PENDING_ABH", "net_payable": 11730.00, "created_at": "2025-04-28T11:05:44Z" }
Error Responses	409 Conflict: duplicate bill (ERR-UT-01-003). Response includes existing_bill_id. 422 Unprocessable: field validation failures. 401, 403, and 500 follow the same pattern as /requests.

Notes	net_payable is calculated server-side as bill_amount plus advance_tax. Any client-submitted net_payable value is ignored.

8.3 Oracle Financial Integration Endpoints

The following outbound calls are made by i-Property to Oracle Financial. Full payload schemas, field mappings, authentication mechanism, and retry policy are specified in IDD-001 Section 3.

POST [OF] /payment-requests | Transmit approved monthly payment batch to Oracle Financial	
Request Headers	Authorization: ApiKey {OF_API_KEY} (stored in a server-side environment variable, never in the client) Content-Type: application/json X-Source-System: I-PROPERTY X-Idempotency-Key: {monthly_summary_id}
Request Body	{ "batch_reference": "PAY-BATCH-2025-04", "generated_by": "I-PROPERTY", "approved_at": "2025-04-30T06:00:00Z", "payments": [{ "landlord_id": "LL-000142", "agreement_id": "AGR-2025-000031", "period": "04/2025", "gross_amount": 75000.00, "tax_deduction": 11250.00, "net_payable": 63750.00, "bank_account": "36ALFA0021000005854802", "payment_reference": "IPROP-REQ-2025-000142-04-2025" }] }
201 Response	HTTP 201 Created (from Oracle Financial). { "of_batch_id": "OF-BATCH-20250430-0042", "status": "QUEUED", "payment_count": 1 } i-Property stores the of_batch_id against the monthly_payment_summary record.
Error Responses	OF 400: schema validation failure. Error is logged. System Admin is alerted. Finance is notified. OF 401 or 403: authentication failure. System Admin is alerted. Retry is not automatic and requires a credential check. OF 5xx: server error. Retry 3 times at 10-minute intervals with exponential backoff. If still failing, System Admin and Finance are alerted with the batch_reference. Network timeout beyond 30 seconds is treated as a 5xx error and follows the same retry policy.
Notes	The X-Idempotency-Key prevents duplicate payment batches if i-Property retries. Bank account data is transmitted encrypted via TLS and is not logged in application logs.

POST [OF] /expense-accruals | Push expense accrual entries for ledger recording	
Request Headers	Authorization: ApiKey {OF_API_KEY} Content-Type: application/json
Request Body	{ "entries": [{ "entry_type": "ACCRUAL", "agreement_id": "AGR-2025-000031", "period": "04/2025", "amount": 75000.00, "cost_centre": "AREA-EAST", "ledger_account": "7100-RENT" }] }

201 Response	HTTP 201 from Oracle Financial with entry reference IDs. i-Property updates the request record with of_accrual_ref.
Error Responses	Same retry and alert policy as the payment-requests endpoint.
Notes	This is triggered when the Admin HO marks a request as COMPLETED. Ledger account codes are to be confirmed with the Finance team and documented in IDD-001 Section 3.3.

9. UI Screen Inventory and Requirements

Detailed wireframes, component specifications, and interaction design are maintained in UI-001. This section defines the functional obligations of each screen, which form the contract between this SRS and UI-001.

Screen ID	Screen Name	Roles	Key Functional Obligations	UI-001 Ref
SCR-01	Login and SSO	All	Redirect to SSO. On success, receive JWT. On failure, show lockout state after 5 attempts.	UI-001 2.1
SCR-02	Dashboard	All (role-filtered)	Show pending approvals count badge, expiry alert count, and quick-action links. Role-specific widgets are shown based on the user's role. All counts reflect live DB state.	UI-001 2.2
SCR-03	New Request Form	Area Officer	Multi-section form with Landlord lookup, Property lookup filtered by landlord, agreement details, and document upload. All mandatory fields are marked with an asterisk. Inline validation on blur. Discard prompt on navigation (FR-NR-05).	UI-001 3.1
SCR-04	Approval Queue	ABH, Director, Admin HO	Paginated list of pending requests with columns for Reference ID, Type, Area, Landlord Name, Submitted Date, and Days Pending highlighted red if SLA is breached. Clicking a row navigates to SCR-05. Bulk approve is not permitted.	UI-001 3.2
SCR-05	Request Detail	All roles	Full request data with editable fields for Admin HO only. Approval history timeline. Document viewer for inline PDF and image preview. Approve, Reject, and Complete action buttons with confirmation modal. Remarks field shown on Reject.	UI-001 3.3
SCR-06	Renewal and Expiry Report	Admin, Area Office, ABH	Table of agreements filtered by expiry look-ahead (configurable, default 90 days). Columns include Agreement ID, Landlord, Area, Expiry Date, and Days Remaining colour-coded as red below 30, amber 30 to 60, and green 60 to 90. An Initiate Renewal button is available per row.	UI-001 3.4

Screen ID	Screen Name	Roles	Key Functional Obligations	UI-001 Ref
SCR-07	Vacated Request Form	Area Admin	Captures termination date (date picker, not in the past), reason (text, minimum 10 characters), and document upload (required). Same discard prompt as SCR-03.	UI-001 3.5
SCR-08	Utility Bills Screen	Area Officer	Search bar by Agreement ID or Landlord ID, results list with connection type. Bill entry form includes bill type dropdown, period picker in MM/YYYY format, and numeric inputs for amounts. Inline duplicate warning shown before submit.	UI-001 4.1
SCR-09	Monthly Payment Summary	ABH/GM and Finance	ABH view: editable grid with stop-payment toggle per row requiring a reason, and an Approve button with confirmation modal. Finance view: read-only with Excel and PDF download buttons. Stopped rows are visually distinguished.	UI-001 5.1
SCR-10	Reports Module	Admin, ABH, Finance	Report type selector with filter panel for area, date range, status, and landlord. Export to Excel and PDF. All filters apply before export. Table preview shows a maximum of 100 rows; full data is included in the export.	UI-001 6.1
SCR-11	Print Queue	Area Admin	Lists completed requests with the agreement document and Approval Summary PDF. A Download Combined Package button merges documents into a single PDF for printing.	UI-001 6.2
SCR-12	User Management	System Admin	Create, edit, and deactivate users. Role assignment with multi-role not permitted. Area mapping. Password reset trigger. Audit log access button.	UI-001 7.1
SCR-13	Notification Configuration	System Admin	Edit HTML templates for all 14 notification events. Configure SLA breach thresholds per approval stage. Test Send button sends a copy to the System Admin email. Template variables are listed per template.	UI-001 7.2

9.1 Global UI Requirements

- All mandatory fields have a red asterisk suffix on the label. Any unfilled mandatory field submitted highlights with a red border and a message shown below.
- All API error responses display the user-facing message from the error.message field. Error codes, stack traces, and internal IDs are never shown in the UI.
- Any action that calls an API shows a loading spinner on the action button. The button is disabled during the request to prevent double submission.
- Session timeout: a modal appears 28 minutes after the last activity warning the user that their session will expire in 2 minutes. At 30 minutes the user is auto-logged out and redirected to SCR-01.
- All tables default sort by created_at descending. Column headers are clickable to toggle sort. Sort state is preserved on page refresh using URL parameters.

- File upload shows the filename, file size, and a remove button after selection. A progress bar is shown during upload. Errors are shown inline if type or size validation fails.
- Confirmation modals are required for all destructive or irreversible actions including Approve, Reject, Stop Payment, and Discard. The modal shows a title, an explanation sentence, and two buttons: a primary action button and a Cancel button.

10. Field Definitions

Field types and constraints defined here represent the application-layer contract. Database column types including indexes and foreign keys are elaborated in SDD-001 Section 4. Error messages shown here are the exact strings the UI must display.

10.1 New Rental Request, Agreement Fields

Field Name	DB Type	Max Length	Mandatory	Validation Rule	Error Message	Notes
reference_id	VARCHAR	20	Auto	System-generated; format REQ-YYYY-NNNNNN; unique constraint in DB	None	Read-only; generated after submission
landlord_id	VARCHAR	20	Yes	Must exist in landlord_profile with status ACTIVE	Landlord ID does not exist or is inactive	Typeahead lookup on SCR-03
property_id	VARCHAR	20	Yes	Must exist in property_profile linked to landlord_id with status ACTIVE	Property not associated with selected Landlord	Filtered by landlord_id selection
area_code	VARCHAR	20	Yes	Must exist in areas table; must be in user's allowed areas	You do not have permission for this area	Dropdown filtered by user's areas
agreement_type	ENUM		Yes	Values: New, Renewal, Transfer	Please select an agreement type	
start_date	DATE		Yes	Not in the past; format DD/MM/YYYY	Start date cannot be in the past	Date picker; past dates disabled
end_date	DATE		Yes	Strictly after start_date	End date must be after Start date	Validates after start_date is selected
rental_amount	DECIMAL	15,2	Yes	Range 1.00 to 99,999,999.99; no negative; numeric only	Amount must be between 1.00 and 99,999,999.99	Formatted with thousand

Field Name	DB Type	Max Length	Mandatory	Validation Rule	Error Message	Notes
						separators in UI
tax_rate	DECIMAL	5,2	Yes	Range 0.00 to 100.00; default loaded from system tax configuration	Tax rate must be between 0 and 100	Pre-filled; editable
escalation_clause	TEXT	500	No	Free text; maximum 500 characters	Maximum 500 characters	Character counter shown in UI
remarks	TEXT	500	No	General remarks; mandatory on REJECT with minimum 1 character enforced at approval step	Remarks are required to reject	Shown and required only on Reject action
document_attachments	FILE[]	10 MB	No	Types: PDF, JPG, PNG; maximum 10 MB per file; maximum 5 files per submission	Only PDF, JPG, PNG permitted or Exceeds 10 MB	Multiple upload; each file validated individually
status	ENUM		Auto	Values: DRAFT, PENDING_ABH, PENDING_DIRECTOR, PENDING_ADMIN_HO, COMPLETED, REJECTED, CLOSED	None	System-controlled; read-only in UI
created_by	VARCHAR	36	Auto	User ID from JWT; cannot be set manually	None	Derived from session context
created_at	TIMESTAMP		Auto	UTC timestamp set by DB on INSERT; displayed in PST in UI	None	
version	INTEGER		Auto	Optimistic lock counter; incremented on each UPDATE; default 1	None	Included in PATCH requests to prevent concurrent edit conflicts

10.2 Utility Bill Fields

Field Name	DB Type	Max Length	Mandatory	Validation Rule	Error Message	Notes
utility_bill_id	VARCHAR	20	Auto	Format UTIL-YYYY-NNNNNN; unique constraint in DB	None	Auto-generated
agreement_id	VARCHAR	20	Yes	FK to agreements; agreement.status must be ACTIVE	Agreement is not active or does not exist	Lookup with search
bill_type	ENUM		Yes	Values: ELECTRICITY, GAS, WATER, OTHER	Please select a bill type	Dropdown
billing_period	VARCHAR	7	Yes	Format MM/YYYY; not in the future; month 1 to 12, year within last 5 years	Billing period cannot be in the future	Month and year picker
units_consumed	DECIMAL	10,2	Yes	Minimum 0; no negative; numeric only	Units consumed cannot be negative	
bill_amount	DECIMAL	15,2	Yes	Minimum 0.01; no negative	Bill amount must be greater than zero	
advance_tax	DECIMAL	15,2	No	Minimum 0; default 0; no negative	Advance tax cannot be negative	
net_payable	DECIMAL	15,2	Calculated	Server-calculated as bill_amount plus advance_tax. Any client-submitted value is ignored.	None	Displayed read-only after submission
bill_document	FILE	10 MB	Yes	Types: PDF, JPG, PNG; maximum 10 MB; exactly 1 file required	Please upload the bill document	
status	ENUM		Auto	Values: PENDING_ABH, APPROVED, REJECTED	None	System-controlled

10.3 Landlord Profile Fields

Field Name	DB Type	Max Length	Mandatory	Validation Rule	Error Message	Notes
landlord_id	VARCHAR	20	Auto	Format LL-NNNNNN; unique constraint in DB	None	Auto-generated on profile creation
full_name	VARCHAR	100	Yes	Minimum 2 characters; letters, spaces, hyphens, and periods only; no numbers	Name must be at least 2 characters	
state_id_number	VARCHAR	20	Yes	Alphanumeric; no single standard format; unique per user recommended	Invalid ID format	No strict mask enforced; flexible entry due to varying state formats
contact_number	VARCHAR	15	Yes	US phone number (mobile or landline); pattern (XXX) XXX-XXXX or XXX-XXX-XXXX; 10 digits	Invalid contact number format	Normalize to 10-digit numeric for storage
email_address	VARCHAR	100	No	RFC 5322 format; if provided, must be unique in landlord_profile	Invalid email address format	
mailing_address	TEXT	255	Yes	Minimum 10 characters; free text	Please enter a complete mailing address	
bank_name	VARCHAR	100	Yes	Minimum 2 characters; free text	Please enter the bank name	
bank_account_no	VARCHAR	30	Yes	Numeric only; minimum 10 digits; stored AES-256 encrypted at rest; displayed masked with last 4 digits visible	Invalid bank account number	Never logged; masked in all API responses
iban	VARCHAR	34	No	2 check digits plus 4 bank code plus 16 account digits; IBAN checksum validated server-side	Invalid IBAN	

Field Name	DB Type	Max Length	Mandatory	Validation Rule	Error Message	Notes
tax_registration	ENUM		Yes	Values: REGISTERED, UNREGISTERED, EXEMPT	Please select tax registration status	
status	ENUM		Yes	Values: ACTIVE, INACTIVE; default ACTIVE	None	INACTIVE blocks new agreements

11. Non-Functional Requirements

Each NFR below includes the specification, the verification method, and the document that owns the implementation detail.

NFR-ID	Category	Specification	Verification Method	Owner Doc	Priority
NFR-01	Performance	Average API response time is 500ms or less at 50 concurrent users. Average page load (TTI) is 3 seconds or less. The system handles 100 concurrent users with no more than 20% response time increase. Report export of 10,000 rows completes within 30 seconds.	JMeter load test with 100 virtual users sustained for 30 minutes. Measure P95 response time per endpoint. Test documented in TP-001 Section 8.	DP-001	High
NFR-02	Availability	99.5% uptime during business hours 08:00 to 20:00 PST Monday to Saturday. RTO is 1 business day. RPO is 24 hours. Planned maintenance window is Sundays 02:00 to 06:00 PST.	Uptime monitoring via Nagios or Zabbix (configured in DP-001 Section 4). Monthly uptime report reviewed by IT Head.	DP-001	High
NFR-03	Authentication	SSO or LDAP via OAuth 2.0 or SAML 2.0. JWT expiry: 8 hours. Refresh token: 24 hours. Session inactivity timeout: 30 minutes with 2-minute warning. Account lockout after 5 consecutive failures with unlock by Admin or 30-minute auto-reset. All API endpoints return 401	Penetration test: attempt API calls without and with expired JWT and verify 401. Functional test: trigger lockout after 5 failures. Session timeout test: leave session idle 32 minutes and verify	SAD-001	Critical

NFR-ID	Category	Specification	Verification Method	Owner Doc	Priority
		within 50ms for invalid or missing tokens.	auto-logout and redirect.		
NFR-04	Authorisation (RBAC)	RBAC is enforced at the API endpoint level and not only in the UI. Area Officers from one area cannot access data from another. Finance cannot POST to /requests or /utility-bills. All permission checks are logged in audit_log.	Role-based test matrix: for each of the 7 roles, attempt access to each of the 13 screen endpoints and verify the correct allow or deny response. Cross-area test and test cases documented in TP-001 Section 5.	SAD-001	Critical
NFR-05	Data Security	TLS 1.2 minimum for all data in transit; TLS 1.3 preferred. AES-256 encryption at rest for bank_account_no and IBAN. bank_account_no is never returned in full in any API response (masked to last 4 digits). No sensitive field values in server logs or client-side storage. HTTP requests are redirected to HTTPS.	SSL certificate audit to verify TLS version and cipher suite. DB column audit to verify encrypted columns. API response audit to verify bank_account_no is masked in all GET /requests responses. Log audit to verify no bank account data appears in application logs.	SAD-001	Critical
NFR-06	Audit Trail	All INSERT, UPDATE, and DELETE operations on core tables are logged with user_id, ip_address, timestamp_utc, and old and new values as JSON. The audit_log table permits no UPDATE or DELETE at the DB level. Audit logs are retained for 3 years active and 7 years archived. System Admin can search by user, date range, table, and record_id.	Create a request and approve it. Verify audit_log contains both INSERT and UPDATE entries with correct before and after JSON. Attempt DELETE on audit_log as the application DB user and verify permission denied. View audit log via SCR-12 and verify search and filter functions.	SAD-001	High
NFR-07	Browser Compatibility	Supported: Chrome (current minus 2), Firefox (current minus 2), Edge (current minus 2). Minimum legacy: Internet Explorer 11 (degraded but functional). Mobile browsers are not in scope for Phase 1.	Cross-browser automated regression suite using Selenium or Playwright covering all 13 screens. Run against Chrome, Firefox, and Edge on Windows 10 and 11. IE11 smoke test: submit request, approve, and export	TP-001	Medium

NFR-ID	Category	Specification	Verification Method	Owner Doc	Priority
			report with no blocking errors.		
NFR-08	Usability	Core workflows including New Request, Renewal, and Utility Bill are completable in 10 screen interactions or fewer. All form fields display an inline validation message within 300ms of the blur event. All user-facing error messages use plain language with no technical codes or stack traces. WCAG 2.1 AA accessibility is the target (to be confirmed in UI-001).	Walkthrough test: count screen interactions for New Request happy path and verify it is 10 or fewer. Error message review: QA to verify all 422 error messages in TP-001 Section 7 use plain language. Blur validation timing test: measure validation message appearance delay.	UI-001	Medium
NFR-09	Scalability	Architecture supports 150 concurrent users (50% growth) without code changes. Horizontal scaling is enabled via a stateless application tier using JWT-based sessions. DB connection pooling is configured with a minimum of 10 and a maximum of 50 connections.	Architecture review in SAD-001 to verify stateless design. Load test at 150 virtual users to verify no more than 50% degradation from the 100-user baseline.	SAD-001, DP-001	Medium
NFR-10	Data Backup	Full database backup daily at 02:00 PST. Incremental backup every 6 hours. Backup retention of 90 days. Restore drill conducted quarterly with documented RTO verification. Backup storage is offsite and separate from the primary database server.	Review backup schedule configuration in DP-001. Quarterly restore drill: restore the latest backup to the test environment, verify data integrity, document the time taken, and compare against the RTO of 1 business day.	DP-001	High

12. Notification Event Specifications

All notifications are delivered asynchronously via the notification job queue through the SMTP relay. On first delivery failure, retry occurs 3 times at 5-minute intervals. If all retries fail: the failure is logged to the notification_failures table, the System Administrator is alerted, and the triggering transaction is not blocked.

NE-ID	Trigger	Recipients	Subject Line	Required Body Fields	Delivery SLA	Priority
NE-01	New request submitted (POST /requests success)	ABH mapped to request area	[Action Required] New Rental Request: {reference_id}	reference_id, landlord_name, property_address, area_name, submission_date, direct_approval_url	2 min	High
NE-02	ABH approves, status becomes PENDING_DIRECTOR	Director Office users	[Action Required] Rental Request Approved by ABH: {reference_id}	reference_id, landlord_name, abh_name, approval_date, direct_approval_url	2 min	High
NE-03	ABH rejects, status becomes REJECTED	Area Officer who submitted	[Update] Rental Request Rejected: {reference_id}	reference_id, landlord_name, rejected_by, rejection_date, rejection_remarks, new_request_url	2 min	High
NE-04	Director approves, status becomes PENDING_ADMIN_HO	Admin Head Office users	[Action Required] Request Approved by Director: {reference_id}	reference_id, landlord_name, director_name, approval_date, direct_review_url	2 min	High
NE-05	Director rejects, status becomes REJECTED	ABH and originating Area Officer	[Update] Request Rejected by Director: {reference_id}	reference_id, landlord_name, rejected_by, rejection_remarks	2 min	High
NE-06	Admin HO completes, status becomes COMPLETED	Area Admin and Finance simultaneously	[Action Required] Agreement Ready: {reference_id}	reference_id, agreement_id, landlord_name, property_address, rental_amount, tax_rate, effective_date, approval_summary_url	2 min	High
NE-07	Monthly payment summary generated by scheduler	All ABH and GM users	[Action Required] Monthly Payment Summary Ready: {period}	period, total_agreements, total_gross_amount, total_net_payable, summary_review_url	By 08:05 PST on generation day	High
NE-08	ABH approves	Finance Department	[Action] Approved Payment	period, approved_by, approval_date,	2 min	High

NE-ID	Trigger	Recipients	Subject Line	Required Body Fields	Delivery SLA	Priority
	payment summary		Summary: {period}	payment_count, total_net_payable, download_url (Excel and PDF)		
NE-09	Weekly scheduled job, Mondays at 08:00 PST	Admin and ABH	[Weekly Report] Agreements Expiring Within {n} Days	report_period, expiry_count, agreements_list with ID and landlord and expiry date and days remaining, renewal_initiation_url	By 08:10 PST Monday	Medium
NE-10	Director approves vacated, payment_stopped set to true	Finance Department (Urgent)	[URGENT] Stop Payment: Agreement {agreement_id}	agreement_id, landlord_id, landlord_name, bank_account (masked), termination_date, last_payment_date, action required to stop all future payments from the specified date	2 min	Critical
NE-11	Utility bill submitted (POST /utility-bills success)	ABH for agreement area	[Action Required] Utility Bill Pending Approval: {utility_bill_id}	utility_bill_id, agreement_id, landlord_name, bill_type, billing_period, net_payable, bill_document_url, approval_url	2 min	High
NE-12	ABH approves utility bill	Finance Department	[Action] Utility Bill Approved for Payment: {utility_bill_id}	utility_bill_id, agreement_id, landlord_name, bill_type, billing_period, net_payable, payment_due_date	2 min	High
NE-13	Stop-payment flag set on landlord in summary	Finance and ABH	[Alert] Payment Stopped: Landlord {landlord_id}	landlord_id, landlord_name, period, stop_reason, stopped_by	2 min	High
NE-14	Approval SLA breached (pending beyond configured threshold per stage)	System Admin and relevant manager	[SLA Breach] Approval Pending: {reference_id}	reference_id, current_stage, assigned_to, submitted_at, hours_pending, sla_threshold_hours	5 min of breach	Medium

13. Requirements Traceability Matrix

The full RTM is maintained in TRM-001. The preview below maps each BRD requirement to the corresponding SRS functional requirement, processing logic section, UI screen, API endpoint, and test case reference.

BRD Ref	Requirement Summary	SRS FR	SRS Section	API Endpoint	UI Screen	TC Ref
FR-NR-01	Auto-generate reference ID	FR-NR-01	6.1	POST /requests	SCR-03	TC-001-A, B, D
FR-NR-02	Email ABH on submission	FR-NR-01	6.1	Async NE-01	None	TC-001-A
FR-NR-03	Rejection requires remarks	FR-NR-02	6.2	PATCH /requests/{id}/action	SCR-04, SCR-05	TC-002-B
FR-NR-04	Full approval routing ABH to Director to Admin HO	FR-NR-02, FR-NR-03	6.2	PATCH /requests/{id}/action	SCR-04, SCR-05	TC-002-A, TC-003-A
FR-NR-05	Area Admin and Finance notified on complete	FR-NR-03	6.2	Async NE-06	None	TC-003-B, C
FR-NR-08	Approval summary PDF generation	FR-NR-04	6.3	Internal PDF engine	SCR-11	TC-004-A, B, C
FR-NR-12	Discard unsaved changes	FR-NR-05	6.4	Browser beforeunload and router guard	SCR-03, SCR-07, SCR-08	TC-005-A to E
FR-NR-09	Monthly payment summary	FR-NR-06	6.5	POST /payment-summary/generate	SCR-09	TC-006-A to E
FR-VT-02	Stop payment on vacated approval	FR-VT-01	6.6	PATCH /requests/{id}/action	SCR-05	TC-007-A to D
FR-UT-01 to 04	Utility bill with duplicate check	FR-UT-01	6.7	POST /utility-bills	SCR-08	TC-008-A to E
NFR-01 to 10	All non-functional requirements	NFR-01 to 10	Section 11	N/A, architectural	N/A	TP-001 Sections 5 to 9
Integration OF	Oracle Financial payment batch	FR-NR-06	Section 8	POST [OF]/payment-requests	None	IDD-001 integration tests
Integration OF	Oracle Financial expense accruals	FR-NR-03	Section 8	POST [OF]/expense-accruals	None	IDD-001 integration tests

14. Document Approval and Sign-Off

Changes after sign-off require a formal Change Request submitted to IT-PMO. The version number will be incremented accordingly.

Name and Role	Department	Date	Signature
Business Analyst, IT-PMO	IT-PMO		
Head of Enterprise Architecture	IT-PMO		
Manager Real Estate	Admin		
Senior Manager Finance	Finance		
Head Business Systems	IT		
Head of Internal Audit	Audit		
Head of Legal	Legal		

By signing this document, stakeholders confirm that the requirements described herein accurately represent the technical and business needs and approve this document as the baseline for system development and testing.